Appendix: Artifact Description

Artifact Description (AD)

I. Overview of Contributions and Artifacts

A. Paper's Main Contributions

- C_1 A deterministic adapter that materializes generator-facing drug, regimen, and condition views from a pinned Unified Oncology Treatment Database (UOTD) release, with source hashes, provenance, and a fail-closed regimen projection audit.
- C_2 A five-category, 20-subcategory oncology-medication extraction taxonomy with fixed quotas and 347 versioned clinical-text templates.
- C_3 A standard-library-only generation and validation pipeline that produces a deterministic 2,000-row benchmark, exact evidence offsets, grounded annotations, and a template-disjoint 1,600/400 train/test split.
- C_4 Dataset interfaces for medication recognition and normalization, prescribing-attribute extraction, contextual interpretation, and regimen resolution, accompanied by category-level profiles and release manifests.

B. Computational Artifacts

- A_1 OncoRx-Bench reproducibility bundle. Generator, configuration, schemas, UOTD adapter, checked-in knowledge views, strict validator, split/profile utilities, regression tests, and manifests. Repository/revision: https://github.com/shivanna-ornl/oncorx-bench/tree/reproducible-v1. No archival DOI is assigned in this draft.
- A_2 OncoRx-Bench dataset release. Canonical and split JSONL and CSV files, validation report, dataset profile, template assignments, and release/split manifests. Dataset revision: https://huggingface.co/datasets/AbhishekShivanna/oncorx-bench/tree/842efd033f21d78b11ee05e6031215e46a440947. Archival DOI: not assigned.

The artifact locations above identify immutable public revisions. The main paper keeps identity-bearing artifact citations anonymous during review; this separately submitted AD records the exact revisions needed for reproduction. The upstream UOTD input is pinned at commit e4ba3722b5505cfa587b30032b8896d86baf8092.

ID	Supports	Related paper elements
A_1	C_1–C_3	Knowledge-grounded generation pipeline figure; source-role, contracts, taxonomy, controls, and software-module tables; integrity results.
A_2	C_2–C_4	Dataset-composition, category-profile, representative-instance, and benchmark-interface tables.

II. Artifact Identification

A. Computational Artifact A_1

Relation To Contributions

Artifact A_1 operationalizes C_1–C_3: its UOTD adapter establishes knowledge lineage, its configuration and templates instantiate the taxonomy, and its release driver connects generation, validation, splitting, profiling, and byte comparison. It supports reproducibility claims, not clinical correctness.

Expected Results

A successful check regenerates 2,000 records byte-for-byte. The knowledge views contain 241 drug rows, 492 regimen rows, 804 condition associations, and 2,151 projection-audit decisions. Validation reports zero schema, quota, identifier, grounding, offset, evidence, signature, placeholder, or duplicate errors. The 1,600/400 split has no shared source template; the canonical JSONL SHA-256 is 97a6c2a22eacf0d68e1762051fbe57b9 43f24c032a014eadd64f7e36aecf952f; and all 24 tests pass. These outcomes establish artifact identity and structural integrity, not clinical adjudication, real-note generalization, or treatment safety.

Expected Reproduction Time (in Minutes)

Budget 2–5 minutes for setup, less than 1 minute for release checking, and less than 1 minute for tests and manifest review. On the verification host, these steps took 0.93 and 3.14 seconds, respectively. A network-dependent UOTD view rebuild may require another 1–5 minutes.

Artifact Setup (incl. Inputs)

Hardware: A 64-bit CPU, 1 GiB available memory, and 100 MiB temporary storage suffice. No accelerator, cluster, interconnect, or privileged service is required; network access is used only by the optional UOTD download check.

Software: The frozen environment is CPython 3.9.6 and A_1 at https://github.com/shivanna-ornl/oncorx-bench/tree/reproducible-v1. All runtime paths use only the standard library; requirements.txt declares no external package. A POSIX-like shell is optional.

Datasets / Inputs: The normative inputs are drug_table.csv, regimen_table.csv, and Conditions_And_Regimens.csv under data/knowledge. Provenance, hashes, and projection decisions are checked in. The optional adapter check uses UOTD commit e4ba3722b5505cfa587b30032b8896d86baf8092. The full synonym inventory is not republished, and the derived views do not supersede upstream terms.

Installation and Deployment: Clone or unpack A_1 and select CPython 3.9.6. No compilation, container, database, credentials, or service is needed. The repository currently has no source-code license; this must be resolved or retained explicitly as an archival reuse limitation.

Artifact Execution

The workflow and dependencies are:

- T_1 Verify the checked-in knowledge-view and provenance hashes.
- T_2 Using random seed 42, instantiate the 20 configured subcategories and 347 templates to produce exactly 2,000 JSONL records.
- T_3 Run strict schema, grounding, evidence, quota, and duplicate validation; any error terminates the workflow with a nonzero status.
- T_4 Construct template-grouped 80/20 splits and recompute exports, profiles, LaTeX statistics, and manifests.
- T_5 Byte-compare every regenerated output with the archive.

Thus, $T_1 \rightarrow T_2 \rightarrow T_3 \rightarrow T_4 \rightarrow T_5$. The single entry point is python3 scripts/reproduce_release.py –check; the accompanying test entry point is python3 -m unittest discover -s tests -v. Because the pipeline is deterministic, no statistical repetitions are used; tests also check cross-process byte identity under different PYTHONHASHSEEDs.

Artifact Analysis (incl. Outputs)

Inspect the validation, split, profile, and release JSON manifests. A pass has zero errors, exact quotas, zero template overlap, and matching SHA-256 values. The profile generates paper/generated_stats.tex, tying paper tables to the canonical JSONL.

B. Computational Artifact A_2

Relation To Contributions

Artifact A_2 supports C_2 and C_4 and is the output of C_3. Its annotations, partitions, and profile underlie the paper's dataset tables and examples. No LLM experiment is reported, so it reproduces the resource and descriptive statistics, not model results.

Expected Results

Expect 2,000 synthetic rows in five categories and 20 subcategories, 3,525 drug objects, 1,286 populated prescribing-information objects, 444 regimen objects, 170 unique drugs, 176 unique regimens, and 347 templates. Full/train/ test counts are 2,000/1,600/400, validation has zero errors, and the canonical hash is 97a6c2a22eacf0d68e1762051fbe57b9 43f24c032a014eadd64f7e36aecf952f.

Expected Reproduction Time (in Minutes)

Budget less than 2 minutes to obtain the 6.5 MiB bundle and 1–2 minutes to check manifests, profiles, and examples. No model execution is involved.

Artifact Setup (incl. Inputs)

Hardware: No specialized hardware is required beyond storage for a 6.5 MiB text bundle.

Software: Any JSONL/CSV reader suffices. CPython 3.9.6 and A_1 are needed only for validation or regeneration; no Hugging Face package is required.

Datasets / Inputs: A_2 at https://huggingface.co/datasets/AbhishekShivanna/oncorx-bench/tree/842efd033f21d78b11ee05e6031215e46a440947contains canonical and split JSONL/CSV plus manifests and profiles. Its text is synthetic and contains no patient records. It is not clinically adjudicated or intended for patient care; redistribution remains subject to the notices and upstream terms in A_1.

Installation and Deployment: Download the immutable A_2 revision; static inspection needs no deployment. For regeneration, use the matching A_1 revision recorded in the release manifest.

Artifact Execution

The dataset-level workflow is $D_1 \rightarrow D_2 \rightarrow D_3$:

- D_1 Verify the downloaded files against the SHA-256 values in release_manifest.json.
- D_2 Confirm row counts, exhaustive split IDs, and zero template overlap.
- D_3 Compare the profile and paper-example IDs, or invoke A_1 for strict validation and regeneration.

Fixed parameters are the 2,000-row quotas, seed 42, nested schema, and template-grouped subcategory-stratified split. No repetitions are applicable.

Artifact Analysis (incl. Outputs)

Compare manifests and the profile with the expected values and paper tables. Checksums establish identity; validation and split checks establish structural consistency, not clinical validity or transfer to real notes.